

proof of (2):

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^H \gamma^t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=t}^H \gamma^{t'-t} r(s_{t'}, a_{t'}) - b(s_t) \right) \right]$$

This form shows the policy gradient as a sum over actions, each scaled by an advantage estimate: a discounted future reward sum minus a baseline.

Maximize

We want to maximize expected return:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)], \quad R(\tau) = \sum_{t=0}^H \gamma^t r(s_t, a_t), \quad \tau = (s_0, a_0, \dots, s_H, a_H).$$

Use the log-derivative trick

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)] = \mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log p_{\pi_{\theta}}(\tau) R(\tau)].$$

proof of the trick:

$$\nabla_{\theta} p(\tau; \theta) = p(\tau; \theta) \nabla_{\theta} \log p(\tau; \theta) \tag{1}$$

$$\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)] = \nabla_{\theta} \int p_{\pi_{\theta}}(\tau) R(\tau) d\tau \tag{2}$$

$$= \int \nabla_{\theta} p_{\pi_{\theta}}(\tau) R(\tau) d\tau \tag{3}$$

$$= \int p_{\pi_{\theta}}(\tau) \nabla_{\theta} \log p_{\pi_{\theta}}(\tau) R(\tau) d\tau \tag{4}$$

$$= \mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log p_{\pi_{\theta}}(\tau) R(\tau)] \tag{5}$$

$$= \nabla_{\theta} J(\theta). \tag{6}$$

Decompose $\log p_{\pi_{\theta}}(\tau)$:

$$p(\tau) = \rho_0(s_0) \prod_{t=0}^H \pi_{\theta}(a_t | s_t) P(s_{t+1} | s_t, a_t),$$

so

$$\nabla_{\theta} \log p_{\pi_{\theta}}(\tau) = \sum_{t=0}^H \nabla_{\theta} \log \pi_{\theta}(a_t | s_t).$$

Plug in:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau} \left[\left(\sum_{t=0}^H \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) R(\tau) \right] = \sum_{t=0}^H \mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau)].$$

Reward-to-Go (Causality)

Only future rewards depend on a_t . Dropping past rewards: (mathematically if you write out the integral terms that do not depend on the integrator factor out and you are left with the gradient of integral of a probability which is the constant 1 and hence zero)

$$\nabla_{\theta} J(\theta) = \sum_{t=0}^H \mathbb{E}_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \underbrace{\sum_{t'=t}^H \gamma^{t'} r(s_{t'}, a_{t'})}_{\text{reward-to-go}} \right].$$

Factor out γ^t :

$$= \sum_{t=0}^H \mathbb{E}_{\tau} \left[\gamma^t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \sum_{t'=t}^H \gamma^{t'-t} r(s_{t'}, a_{t'}) \right].$$

Add a Baseline (As to what a baseline is and why we subtract see pp 329 of Sutton, but here we give the proof)

Subtract a baseline $b(s_t)$:

$$\nabla_{\theta} J(\theta) = \sum_{t=0}^H \mathbb{E}_{\tau} \left[\gamma^t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=t}^H \gamma^{t'-t} r(s_{t'}, a_{t'}) - b(s_t) \right) \right].$$

proof that you can subtract the base so long as it is a function of state alone:

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim p_{\pi_{\theta}}(\tau)} \left[\sum_{t=0}^H \gamma^t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=t}^H \gamma^{t'-t} r(s_{t'}, a_{t'}) - b(s_t) \right) \right]$$

We explicitly verify that the baseline is unbiased:

$$\begin{aligned}
& \mathbb{E}_{\tau \sim p_{\pi_\theta}(\tau)} \left[\sum_{t=0}^H \gamma^t \nabla_\theta \log \pi_\theta(a_t | s_t) b(s_t) \right] \\
&= \sum_{t=0}^H \gamma^t \mathbb{E}_{\tau \sim p_{\pi_\theta}(\tau)} \left[b(s_t) \mathbb{E}_{a_t \sim \pi_\theta(a_t | s_t)} [\nabla_\theta \log \pi_\theta(a_t | s_t) | s_t] \right] \\
&= \sum_{t=0}^H \gamma^t \mathbb{E}_{\tau \sim p_{\pi_\theta}(\tau)} \left[b(s_t) \int_{a_t} \pi_\theta(a_t | s_t) \nabla_\theta \log \pi_\theta(a_t | s_t) da_t \right] \\
&= \sum_{t=0}^H \gamma^t \mathbb{E}_{\tau \sim p_{\pi_\theta}(\tau)} \left[b(s_t) \int_{a_t} \nabla_\theta \pi_\theta(a_t | s_t) da_t \right] \\
&= \sum_{t=0}^H \gamma^t \mathbb{E}_{\tau \sim p_{\pi_\theta}(\tau)} [b(s_t) \cdot 0] \\
&= 0
\end{aligned}$$

In the finite-horizon derivation we write

$$\nabla_\theta = \sum_{t=0}^H \mathbb{E}_{s_t \sim d_t^\pi, a_t \sim \pi_\theta} [\gamma^t \nabla_\theta \log \pi_\theta(a_t | s_t) \hat{A}(s_t, a_t)].$$

Here both the state-distribution d_t^π and the discount weight γ^t are explicitly indexed by t .

When passing to the infinite-horizon form, those two pieces are folded into a single “discounted occupancy” measure

$$d^\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t d_t^\pi(s),$$

so that the gradient can be written as

$$\nabla_\theta J = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a | s) \hat{A}(s, a)].$$

The stray subscript “ t ” on the state distribution in the infinite-horizon equation was simply a leftover from the finite-horizon version. The correct infinite-horizon line should read

$$\nabla_\theta J = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi(s), a \sim \pi_\theta(a | s)} [\nabla_\theta \log \pi_\theta(a | s) \hat{A}(s, a)],$$

with no time index on d^π .

While a simple time-dependent baseline $b_t = \frac{1}{N} \sum_{i=1}^N R_{i,t}$ is often used to reduce variance in Monte Carlo policy gradient estimators, it does not represent

a true value function $V(s_t) = \mathbb{E}[R_t \mid s_t]$. This is because b_t marginalizes over all states s_t encountered at time t rather than conditioning on a specific state. Thus, it captures only the average return under the state visitation distribution $d_t(s)$, not the expected return from a particular state $s_t = s$. Although useful for variance reduction, this approach lacks the precision and generalization capabilities of a learned, state-dependent baseline.